

Rotational Diagonal Kalman Linear Attention: Model, Mamba-3 Lessons, and an Implementation Contract

August 28, 2026

Abstract

Mamba-2 supplies a strong selective state-space baseline, but its SISO state transition is a scalar radial contraction: it cannot rotate a two-dimensional latent subspace. The installed Mamba-3 SISO implementation adds rotation economically by accumulating a bounded angular rate and applying the same rotary transform to queries and keys. Its most useful lesson is therefore not a particular state-space write rule, but a representation: pairwise rotation can be moved out of the recurrent state update and into a phase-rotated query/key basis.

Rotational diagonal Kalman linear attention (RDiag KLA, E2 clock) adopts that representation while retaining KLA’s diagonal Kalman covariance and innovation write. It pair-ties radial transition and process-noise parameters, uses a detached live radial statistic to set a bounded token/head clock, scans phase in a documented negative gauge, and isolates the phase VJP from radial dynamics. This note defines the model, the precise finite-precision implementation contract, and the tests needed for promotion. A step-520 screen after 519 completed optimizer updates accepted the E2 clock design, but the optimized composed implementation still requires its complete post-integration correctness, distributed, soak, and performance qualification; this document does not declare production readiness.

1 Scope and notation

Let B, T, H, K, V, D denote batch size, sequence length, number of heads, key dimension per head, value dimension per head, and model width. The rotary prefix has even width $K_r = 2P$, where P is its number of adjacent pairs. Tensors use the following shapes:

$$\begin{aligned} \mathbf{x} &\in \mathbb{R}^{B \times T \times D}, & \mathbf{q}, \mathbf{k}, \boldsymbol{\alpha}, \boldsymbol{\omega} &\in \mathbb{R}^{B \times T \times H \times K}, \\ \mathbf{v} &\in \mathbb{R}^{B \times T \times H \times V}, & \mathbf{e} &\in \mathbb{R}^{B \times T \times P}, \\ \boldsymbol{\tau} &\in \mathbb{R}^{B \times T \times H}, & \boldsymbol{\delta}, \boldsymbol{\phi} &\in \mathbb{R}^{B \times T \times H \times P}. \end{aligned}$$

The raw angle e_{btj} is shared across heads; the clock τ_{bth} makes the resulting increment head-specific. For an adjacent pair $(2j, 2j + 1)$, define

$$R(\vartheta) = \begin{bmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{bmatrix}, \quad R(\vartheta)^\top R(\psi) = R(\psi - \vartheta). \quad (1)$$

All equations below are per batch element unless a batch index is useful. “SISO” follows the Mamba implementation’s terminology; a head can still carry multiple value channels that share the same scalar state-space coefficients.

2 What Mamba-2 has—and what it lacks

For head h , value channel p , and SSM state coordinate n , the Mamba-2 selective SISO update can be written

$$d_{th} = \text{softplus}(\tilde{d}_{th} + b_h^d), \quad a_{th} = \exp(A_h d_{th}), \quad A_h < 0, \quad (2)$$

$$X_{t,h,p,n} = a_{th} X_{t-1,h,p,n} + d_{th} B_{t,n} u_{t,h,p}, \quad y_{t,h,p} = \sum_n C_{t,n} X_{t,h,p,n} + D_h u_{t,h,p}. \quad (3)$$

The input projection is ordered as $[z, x, B, C, dt]$. It emits one dt row per token and head; B and C are token-dependent state vectors. The factor a_{th} is a scalar radial contraction over the head state, and $d_{th} B_t u_t$ is its discretized injection. There is no angle projection, phase state, adjacent-pair structure, sine/cosine operation, or 2×2 rotation in this path. Mamba-2 therefore supplies selective decay and injection, but not oscillatory transport. A pair of real coordinates within a head shares the same scalar decay, but that factor is $a_{th} I$: it has no skew component, phase state, or mechanism for rotating one coordinate into the other.

3 The installed Mamba-3 SISO rotational path

This section describes the code shipped in the installed `mamba-ssm 2.3.2.post1` package, not an inferred variant. Its input projection is ordered

$$[z, x, B, C, dd_{dt}, dd_A, \text{trap}, \text{angle}]. \quad (4)$$

The dd_{dt} , dd_A , and trapezoid fields each have H rows. In contrast, the angle field has only P_M rows, where P_M is half of the selected half- or full-rotary state prefix. The raw angle rows are expanded across heads. Thus Mamba-3 separates a token/head clock from token/pair angular content:

$$DT_{th} = \text{softplus}(dd_{dt,th} + b_h^d), \quad (5)$$

$$A_{th} = \min\{-\text{softplus}(dd_{A,th}), -A_{\text{floor}}\}, \quad \rho_{th} = \exp(A_{th} DT_{th}), \quad (6)$$

$$\nu_{tj} = \pi \tanh(e_{tj}), \quad (7)$$

$$\Delta\theta_{thj} = DT_{th} \nu_{tj}, \quad (8)$$

$$\theta_{thj} = \theta_{0,hj} + \sum_{s=1}^t \Delta\theta_{shj} \text{pmod} 2\pi. \quad (9)$$

The cumulative phase is positive. For every selected adjacent pair, the SISO kernel applies the same $R(+\theta_{thj})$ to Q_t and K_t . Consequently the same-token dot product is unchanged, whereas a cross-token term obtains the relative rotation

$$(R(\theta_t) q_t)^\top R(\theta_s) k_s = q_t^\top R(\theta_s - \theta_t) k_s. \quad (10)$$

If S_t denotes a state in a fixed physical frame, this is equivalent to the moving-frame recurrence

$$S_t^- = \rho_t R(-\Delta\theta_t) S_{t-1}, \quad (11)$$

pair by pair, with scalar radial decay ρ_t . The identity follows from $R(\theta_t)^\top R(\theta_{t-1}) = R(-\Delta\theta_t)$. It is the key computational idea: no dense rotational state transition is needed when radial coefficients are pair-isotropic.

Mamba-3 additionally uses a trapezoidal SSM write. With

$$\gamma_t = DT_t \text{sigmoid}(\text{trap}_t), \quad \eta_{t+1} = DT_{t+1}(1 - \text{sigmoid}(\text{trap}_{t+1})), \quad (12)$$

the offline kernel scales token t 's rotated key by $\gamma_t + \eta_{t+1}$, while its same-token contribution uses γ_t . At a streaming boundary, the cached previous key/value contributes the missing η_t term. This is a two-endpoint discretization of the Mamba write; it is not the KLA innovation update defined below.

The angle clock is also coupled in backward. The VJP of the positive cumulative phase is a reverse cumulative sum. It produces both

$$\bar{e}_{tj} = \sum_h \overline{\Delta\theta}_{thj} DT_{th} \pi (1 - \tanh^2(e_{tj})), \quad (13)$$

$$\overline{DT}_{th}^{\text{angle}} = \sum_j \overline{\Delta\theta}_{thj} \pi \tanh(e_{tj}). \quad (14)$$

Here the sum over h is the backward of the head expansion of the shared raw angle rows. The combined implementation explicitly adds $\overline{DT}^{\text{angle}}$ to the DT gradient from the radial SSM dynamics. This is a coherent choice for a shared physical discretization clock, but it is intentionally not the RDiag E2 gradient contract.

4 RDiag KLA with the E2 bounded clock

4.1 Pair-tied radial dynamics

Let the radial preactivation be f_{bthk} with bias b_{hk}^d , and let

$$d_{bthk} = \text{softplus}(f_{bthk} + b_{hk}^d). \quad (15)$$

For rotary pair j , define the pair step and tied process noise

$$\bar{d}_{bthj} = \frac{1}{2}(d_{bth,2j} + d_{bth,2j+1}), \quad (16)$$

$$\bar{\omega}_{bthj} = \frac{1}{2}(\omega_{bth,2j}^{\text{in}} + \omega_{bth,2j+1}^{\text{in}}), \quad (17)$$

$$\alpha_{bth,2j} = \alpha_{bth,2j+1} = \exp\left[-\exp(A_h^{\log})\bar{d}_{bthj}\right], \quad (18)$$

$$\omega_{bth,2j} = \omega_{bth,2j+1} = \bar{\omega}_{bthj}. \quad (19)$$

Only the selected prefix is pair-tied; tail lanes retain the diagonal-KLA frontend's original α and ω . Pair tying is not cosmetic. On each pair it makes $\text{diag}(\alpha_t) = \alpha_{tj}I_2$ and the diagonal process covariance $\text{diag}(\omega_t) = \omega_{tj}I_2$, so the *prediction* terms commute with $R(\vartheta)$. It does not force posterior information or the Kalman gain to be pair-isotropic; an observation generally splits them. The diagonal posterior approximation is defined in the current moving frame, as made explicit below.

4.2 Detached live radial clock and persistent anchor

Set the open upper clock bound to $C = 0.1$. At construction, snapshot one FP32 anchor per head,

$$a_h = \text{mean}_k \text{softplus}(b_{hk}^d), \quad 0 < a_h < C. \quad (20)$$

The anchor is a persistent buffer, not a parameter. At each token compute the detached live radial mean

$$m_{bth} = \text{mean}_k \text{softplus}\left(\text{stopgrad}(f_{bthk} + b_{hk}^d)\right), \quad (21)$$

and a learnable token/head residual

$$z_{bth}^{\text{clk}} = \frac{\langle w_h, x_{bt} \rangle}{\sqrt{D}} + \beta_h, \quad (22)$$

evaluated in FP32. The E2 clock is

$$\tau_{bth} = C \text{ sigmoid} \left[\log m_{bth} - \log(C - a_h) + \tanh(z_{bth}^{\text{clk}}) \right] \quad (C = 0.1). \quad (23)$$

The bounded residual is strictly in $(-1, 1)$ and the sigmoid keeps a finite valid clock strictly in $(0, C)$. The implementation does not clamp or use a straight-through estimator: invalid, nonfinite, endpoint, or dead-derivative conditions are made visible to health checks, with invalid clock outputs replaced by NaNs.

Detachment is a model choice, not merely an optimization. The current radial timescale determines the forward phase scale, while the phase loss cannot change f , b^d , $A^{\log}$, α , or ω through this clock. The persistent anchor fixes the initialization reference across save/resume; recomputing it from a trained b^d would change the model after restart.

4.3 Angle, sign convention, and rotating basis

One projection shared across heads produces e_{btj} . The head-specific increment is

$$\delta_{bthj} = \pi \tanh(e_{btj}) \tau_{bth}, \quad |\delta_{bthj}| < 0.1\pi. \quad (24)$$

RDiag uses the negative inclusive phase scan

$$\phi_{bthj} = \phi_{b0,hj} - \sum_{s=1}^t \delta_{bshj}, \quad (25)$$

implemented in FP32 and reduced to $[0, 2\pi)$ at the bounded-clock consumer. The same $R(+\phi_{thj})$ is applied to the corresponding pairs of q_t and k_t . Hence, for $s < t$,

$$(R(\phi_t)q_t)^\top R(\phi_s)k_s = q_t^\top R\left(\sum_{u=s+1}^t \delta_u\right)k_s. \quad (26)$$

Under the column-vector convention of Eq. (1), the negative scan therefore represents $R(+\delta_t)$ transport. Mamba-3’s positive scan represents $R(-\Delta\theta_t)$. This sign is a gauge choice because replacing the zero-initialized angle projection by its negative maps one convention to the other. It is nevertheless an implementation invariant: forward, streaming carry, and the negative suffix-scan VJP must use the same sign.

More explicitly, let $\mathcal{R}_t$ be block diagonal with the pair rotations $R(\phi_{thj})$ and an identity tail. The corresponding physical transition and process covariance are

$$D_t = \mathcal{R}_t^\top \text{diag}(\alpha_t) \mathcal{R}_{t-1}, \quad \Omega_t^{\text{phys}} = \mathcal{R}_t^\top \text{diag}(\omega_t) \mathcal{R}_t. \quad (27)$$

On a tied pair, these reduce to $D_{tj} = \alpha_{tj} R(+\delta_{tj})$ and $\Omega_{tj}^{\text{phys}} = \omega_{tj} I_2$. With moving-frame state $\bar{S}_t = \mathcal{R}_t S_t$ and addresses $\bar{k}_t = \mathcal{R}_t k_t$, $\bar{q}_t = \mathcal{R}_t q_t$, the mean-state recurrence becomes purely radial and can reuse the diagonal-KLA memory kernel.

4.4 Diagonal Kalman gain and innovation memory

Let the phase-rotated addresses be $\tilde{q}_t = \bar{q}_t$ and $\tilde{k}_t = \bar{k}_t$. With posterior diagonal information $c_{t-1,hk} > 0$, observation noise $r_{th} > 0$, and information scale $s_h > 0$, RDiag KLA uses the moving-frame state (written as S_t below for brevity)

$$p_{thk}^- = \frac{\alpha_{thk}^2}{c_{t-1,hk}} + \omega_{thk}, \quad (28)$$

$$\kappa_{thk} = \frac{p_{thk}^- \tilde{k}_{thk}}{r_{th} + \sum_i p_{thi}^- \tilde{k}_{thi}^2}, \quad (29)$$

$$c_{thk} = \frac{1}{p_{thk}^-} + s_h \frac{\tilde{k}_{thk}^2}{r_{th}}. \quad (30)$$

For associative state $S_{th} \in \mathbb{R}^{K \times V}$,

$$S_{th}^- = \text{diag}(\alpha_{th}) S_{t-1,h}, \quad (31)$$

$$S_{th} = S_{th}^- + \kappa_{th} \left(v_{th} - \tilde{k}_{th}^\top S_{th}^- \right)^\top, \quad (32)$$

$$o_{th} = S_{th}^\top \left(K^{-1/2} \tilde{q}_{th} \right). \quad (33)$$

This is the KLA estimator. It is not a Mamba SSM write: κ_t is an adaptive anisotropic Kalman gain and the write is the current innovation. The diagonal information recurrence can be computed by its existing associative Möbius scan, followed by the existing chunked affine memory recurrence. In physical coordinates, c_t represents the model’s diagonal projection of $\mathcal{R}_t P_t^{-1} \mathcal{R}_t^\top$; it is not a claim that rotating an arbitrary anisotropic posterior covariance remains diagonal. In particular, c , κ , k , and q are never pair-tied.

4.5 VJP isolation

The computational graph has two intentionally separate branches:

$$(f, b^d, A^{\log}, \omega^{\text{in}}) \longrightarrow (\alpha, \omega) \longrightarrow \text{Kalman dynamics},$$

$$\text{stopgrad}(f + b^d), a, (W, \beta), e \longrightarrow \tau, \delta, \phi \longrightarrow \text{rotated } q, k.$$

The direct radial backward has no angle input. Conversely, the clock’s m and a arguments are detached, so the phase VJP returns no gradient to the radial gate, radial bias, or $A^{\log}$. Unlike Mamba-3, RDiag does not add an angular-clock contribution to the dynamical radial-step gradient. Among the new phase-specific parameters, gradients reach the angle projection and clock weight/bias; they also return to the shared hidden states through both projections. The isolation claim is specifically that no clock edge reaches the radial parameters.

5 What to take from Mamba-3

The useful transfer from Mamba-3 is structural: encode rotation as a relative query/key phase with a small shared angle projection and a token/head clock. The estimator-specific pieces should remain KLA-specific.

Aspect	Mamba-2 SISO	Mamba-3 SISO	RDiag KLA E2
Transition geometry	Scalar radial decay; no rotation	Radial SSM plus relative pair rotation	Pair-tied radial KLA plus relative pair rotation
Clock	$DT_{th} > 0$	$DT_{th} > 0$ shared by decay and angle	$0 < \tau_{th} < 0.1$, based on detached live radial mean plus residual
Angle rows	None	Token/pair rows shared across heads	Token/pair rows shared across heads; head-specific τ
Phase gauge	None	Positive cumulative phase	Negative cumulative phase; sign documented by Eq. (26)
Write	Euler-like $DT Bx$	Trapezoidal two-endpoint SSM write	Kalman innovation with anisotropic gain
Clock gradient	Only SSM dynamics	Angle dDT is added to dynamical dDT	Phase/radial VJPs intentionally isolated
Covariance	No Kalman posterior covariance	No KLA diagonal information update	Diagonal information in moving frame; pair-isotropic prediction parameters

Concretely, RDiag should take the following from Mamba-3:

1. represent rotation by the same adjacent-pair transform on queries and keys, avoiding a dense recurrent rotation;
2. use a bounded angular rate $\pi \tanh(e)$ and one token/head clock broadcast over pair modes;
3. share the compact angle projection across heads, while permitting head-specific timing;
4. accumulate phase and streaming carry in FP32, with a specified modular reduction contract;
5. treat every dependency of a shared clock explicitly in backward and test cancellation-heavy reductions against the executable reference; and
6. retain half- or full-prefix rotary layouts rather than forcing every key lane to rotate.

It should intentionally *not* copy the following:

1. Mamba-2’s scalar-only transition as a purported rotational mechanism;
2. Mamba-3’s trapezoidal SSM write, because KLA’s innovation update defines a different estimator;
3. Mamba-3’s angle-to- DT gradient coupling, because E2 deliberately uses a detached radial timescale;
4. approximate softplus, trigonometric, or reassociated reduction sequences without proving the required finite-precision parity;
5. independent radial coefficients inside a rotated pair while claiming a literal scalar-damped rotation; they instead define a more general rotated anisotropic contraction; or
6. hard clock clamps or straight-through boundary gradients, which conceal saturation instead of making it observable.

6 Implementation contract

6.1 Layout and forward pipeline

The production-oriented route is restricted to contiguous CUDA tensors, BF16 projected $q, k, v, f_{\text{raw}}, z_{\text{raw}}, y_{\text{raw}}$, FP32 frontend parameters, equal key/value head counts, and $K \in \{64, 128\}$.

The clock preactivation z^{clk} is a separate FP32 quantity. The rotary fraction is 1/2 or 1, so K_r is positive and even. Adjacent memory lanes $(0, 1), (2, 3), \dots, (K_r - 2, K_r - 1)$ form the rotary pairs; no split-half pairing is permitted.

The forward pipeline is:

1. project q, k, v and the diagonal-KLA radial/noise quantities;
2. compute the standard frontend’s frozen α^{in} and ω^{in} ;
3. in the RDiag direct radial kernel, recompute the rotary prefix using Eq. (19), preserve the tail, and build the log-decay prefix consumed by the chunked memory kernel;
4. form the exact detached FP32 radial mean m described below;
5. keep Wx as the existing FP32 GEMM, then fuse the pointwise clock chain and Eq. (24);
6. perform the bounded negative phase scan in chunks of 64, retaining an FP32 carry;
7. normalize q, k in FP32, round the memory-facing copies as specified, and rotate adjacent pairs by the same phase; and
8. run the diagonal information/gain kernel and the S3 chunked KLA memory recurrence.

The gain-facing key remains FP32. The memory-facing query/key route is rounded through BF16 in the same place as its executable reference before rotation/final rounding; changing that boundary is a numerical model change for parity purposes.

6.2 The precise detached-clock mean

The actual radial dynamics retain the FLA softplus used by the qualified baseline. The detached clock uses a separate precise FP32 copy

$$\text{softplus}_{20}(x) = \begin{cases} x, & x > 20, \\ \log 1p(\exp x), & x \leq 20, \end{cases} \quad (34)$$

evaluated with the precise device intrinsics. To reproduce the eager reference, the optimized path must materialize

$$\text{clock_dt} \in \mathbb{R}^{B \times T \times H \times K} \quad (35)$$

in FP32 and then invoke the framework reduction

$$\text{radial_clock} = \text{torch.mean}(\text{clock_dt}, \text{dim}=-1, \text{out}=\text{radial_clock}). \quad (36)$$

A mathematically equivalent Triton lane sum is not an exact substitute. It changes addition order and was shown to leave small residual differences that can be amplified by long phase scans and cancellation-heavy angle gradients. The precise clock copy is detached; the FLA-softplus radial outputs and their VJP remain untouched. The temporary costs exactly $4BTHK$ bytes. At $B = 4, T = 4096, H = 16$, that is 64 MiB for $K = 64$ and 128 MiB for $K = 128$; at $H = 18, K = 128$ it is 144 MiB. This cost belongs in the end-to-end memory and speed gate rather than being hidden by a numerically different reduction.

6.3 Phase scan and backward order

The phase forward is an inclusive FP32 scan. Each chunk computes $\phi = \text{carry} - \text{cumsum}(\delta)$, wraps valid values to $[0, 2\pi)$, and passes the wrapped final value as the next carry. Before reduction, $|\phi|$ must not exceed 65,536; nonfinite, out-of-contract, or noncanonical residues become sticky NaNs. Its VJP is the global negative suffix scan

$$\bar{\delta}_t = - \sum_{u=t}^T \bar{\phi}_u, \quad (37)$$

with the final-carry cotangent subtracted from every token when present.

For the fused clock/delta backward, operation order is part of the contract. Let $g_{thj} = \bar{\delta}_{thj}$. The eager-compatible order is

$$\bar{\tau}_{th} = \sum_j [g_{thj} (\pi \tanh(e_{tj}))], \quad (38)$$

$$\bar{z}_{th}^{\text{clk}} = \text{mask}_{th}(\bar{\tau}_{th}) \tau_{th}^{\text{raw}} \left(1 - \frac{\tau_{th}^{\text{raw}}}{C}\right) \left(1 - \tanh^2 z_{th}^{\text{clk}}\right), \quad (39)$$

$$\bar{e}_{tj} = \left[\left(\sum_h g_{thj} \tau_{th} \right) \pi \right] (1 - \tanh^2 e_{tj}). \quad (40)$$

Here $\text{mask}_{th}(g)$ is g when the forward validity predicate selected τ_{th}^{raw} and zero when it selected the detached NaN branch. The subsequent multiplication order is retained, so a NaN local derivative remains fail-visible even after a zero cotangent mask. Equation (38) must be evaluated as the displayed broadcast multiply followed by the framework’s sum over P ; pulling π outside the sum is not accepted as “equivalent.” For Eq. (40), multiply by τ , reduce over H , multiply by π , and only then apply the tanh derivative. The validity mask is applied to the incoming clock cotangent before the sigmoid/tanh chain, matching the eager `where(valid, raw_tau, detached_nan)` VJP. After the FP32 head reduction and factor order in Eq. (40), the raw-angle gradient is stored in `angle_raw`’s dtype (BF16 on the production route) before the angle projection VJP. Weight and hidden-state gradients then flow through the ordinary linear operations.

6.4 Persistent state and FSDP

The anchor a_h must satisfy all of the following:

1. FP32, contiguous, shape $[H]$, finite, and strictly inside $(0, 0.1)$;
2. registered as a persistent buffer and present in the model state dictionary;
3. absent from named parameters and optimizer groups;
4. identical across distributed ranks after construction, FSDP wrapping, save, and resume;
5. restored from the checkpoint rather than regenerated from the current radial bias.

FSDP mixed precision must not downcast this buffer. Resume tests must compare its bytes before and after round trips and verify that fresh construction cannot silently overwrite a checkpointed anchor.

6.5 Initialization and health signals

The angle projection weight and bias, and the bounded-clock residual weight and bias, are initialized to zero. Rotation therefore starts at the *pair-tied* diagonal-KLA function: $e = 0$ gives $\delta = 0$ even when τ is nonzero, while the pair averaging in Eq. (19) remains active. This is preferable to initializing the clock at zero, which would place it on a boundary and kill useful clock gradients once angles become live.

The audited bounded RDiag named configurations use $r_{\min} = 0.01$, process-noise floor $\omega_{\min} = q_{\min} = 0$, and `gain_init=iso`. The last setting zeroes the output map of the process-noise projection, so ω starts uniform across key channels; it does *not* initialize ω itself to zero, because the positive softplus remains. These are recipe and initialization choices rather than changes to Eqs. (28)–(33).

At every checked optimizer boundary, diagnostics should aggregate across microbatches, layers, and ranks:

- nonfinite counts for the radial mean, base logit, residual, logit, raw clock, valid clock, derivative, phase, parameters, and gradients;
- exact endpoint and near-boundary counts, using 10^{-4} and $0.1 - 10^{-4}$ as the observed lower and upper near-boundary bands;
- minimum positive clock Jacobian and a near-dead count below 10^{-8} ;
- token variation of the residual projection after bootstrap;
- phase magnitude, delta magnitude, and canonical wrapped-range checks; and
- schema/count agreement across ranks, including exactly one observation per expected microbatch/layer pair.

The kernel itself fails visibly at invalid endpoints. Promotion screens may impose stricter distributional thresholds than the continuous health controller; both classes of gate are needed.

7 Reference pseudocode

The following pseudocode fixes the mathematical order without prescribing launch geometry.

1. Radial and clock preparation.

$$\begin{aligned} x^d &= f + \text{dt_bias}, \\ d^{\text{radial}} &= \text{FLA_softplus}(x^d), \\ \text{clock_dt} &= \text{softplus}_{20}(\text{stopgrad}(x^d)) \quad \text{materialized FP32 } [B, T, H, K], \\ m &= \text{torch.mean}(\text{clock_dt}, \text{dim}=-1, \text{out}=\text{radial_clock}). \end{aligned}$$

Pair-average d^{radial} and ω on the rotary prefix, and construct α from Eq. (19). Do not use `clock_dt` for radial dynamics.

2. Bounded clock and phase.

$$\begin{aligned} z^{\text{clk}} &= \text{linear}(x, W) / \sqrt{D} + \beta, \\ \tau &= 0.1 \text{ sigmoid}(\log m - \log(0.1 - a) + \tanh z^{\text{clk}}), \\ \delta &= \pi \tanh(e) \tau, \\ \phi_t &= \text{wrap}_{2\pi}(\phi_{t-1} - \delta_t). \end{aligned}$$

Reject invalid open-clock or phase-contract values by producing observable NaNs.

3. **Addresses and Kalman recurrence.** Rotate the same adjacent prefix of normalized q and k by $R(\phi)$, preserve tail lanes, then apply Eqs. (28)–(33). The chunked implementation must match this token recurrence under the documented BF16/FP32 boundary.
4. **Backward.** Run Eq. (37); use Eqs. (38)–(40) exactly; return no clock gradient to m or a ; and run the independent radial/Kalman VJPs.

8 Lessons from the E-screen and performance/numerical debugging

An open interval is not a distributional guarantee. An earlier bounded-clock E screen completed 519 optimizer updates and produced no exact endpoints or nonfinite clocks, yet the step-520 evaluator rejected it. In layer 0, the lower-near-boundary fraction was 0.00205326, above the preregistered 10^{-3} threshold; the observed clock range was approximately $[8.65 \times 10^{-5}, 0.09813]$. Thus a smooth sigmoid bound alone did not prevent probability mass from accumulating near a boundary. The design response was the detached live radial base plus a bounded token/head residual, not a harder clamp.

The redesigned E2 screen reached its step-520 evaluation after 519 completed optimizer updates (520 measured gradients), with zero lower and upper near-boundary counts in the reported gate and an observed clock range of approximately $[0.01060, 0.08181]$. That result supports the clock design and justified integration work. It does not prove long-run training quality, kernel parity under every shape, distributed resume correctness, or a production speedup.

Correctness-first composition exposed the actual performance target. At $B = 4, T = 4096, H = 16, K = V = 128, D = 2048$, the unfused rotational-v1 wrapper measured 24.706 ms for combined mixer forward/backward versus 13.735 ms for its frozen control, a $1.799\times$ ratio (79.9% overhead). The identical-input frozen S3 stage was $0.999\times$ its control, localizing the cost to the eager second- dt , pair tying, phase/trigonometry, three rotations, and prefix rebuild. This motivated the direct radial kernel, fused normalization/rotation, and fused clock/delta path while retaining the established gain and S3 kernels. These are historical isolation numbers for the v1 geometry, not a speed claim for the final E2 composition; the exact clock mean’s FP32 scratch and full-layer runtime still require measurement.

A tiny local error can become a large gradient discrepancy. The first optimized clock path reused the FLA radial softplus. Against eager PyTorch, the initial radial-mean discrepancy was only about 6.33×10^{-8} in maximum absolute value and 1.54×10^{-6} in relative L_2 . After the logit, long phase scan, BF16 rotation boundary, and cancellation across heads, the angle-weight gradient showed about 3.81×10^{-3} relative L_2 error in the focused diagnostic. Near-zero reference entries also made elementwise relative metrics misleading, so absolute, normwise, and cancellation-stress gates are all required.

Using Eq. (34) reduced the clock error substantially, but a Triton reduction still did not reproduce eager bitwise. Materializing the FP32 $[B, T, H, K]$ precise values and using the exact framework mean was bitwise identical in the focused $K = 64$ and $K = 128$ experiment while leaving the FLA radial α , log-prefix, and ω outputs bitwise unchanged. Likewise, algebraically reassociated all-Triton backward reductions failed the focused VJP gate; the explicit framework reduction of Eq. (38) and the factor order of Eq. (40) passed it. The lesson is narrow but important: the executable eager graph, including reduction order, is the reference for this mixed-precision path.

9 Invariants and promotion checklist

The following are invariants, not tunable approximations:

1. rotary prefix width is positive and even; pairing is adjacent;
2. α and ω are exactly equal within each selected pair;
3. q and k receive the same phase and same rotation matrix;
4. the phase scan is negative and inclusive, and its backward is a negative suffix scan;
5. m and a are detached FP32 quantities; phase gradients do not enter radial dynamics;
6. the valid clock and its local derivative are finite and strictly positive, with $0 < \tau < 0.1$;
7. the anchor is persistent FP32 checkpoint state, not optimizer state;
8. precise clock softplus plus `torch.mean` is separate from FLA radial softplus;
9. the displayed forward and backward reduction orders are preserved; and
10. the unrotated tail is an exact pass-through of diagonal KLA.

Before production promotion, the following checklist should be closed:

1. pure-PyTorch token recurrence versus optimized forward for zero, half, and full rotation;
2. first-order VJPs for every input/parameter, including adversarial cancellation and partial chunks, at $K = 64$ and $K = 128$;
3. exact checks for radial outputs and detached clock mean where exactness is part of the contract, plus justified tolerances elsewhere;
4. segmented/streaming equivalence, phase carry, wrap-boundary, and sticky-NaN tests;
5. zero-angle equivalence to pair-tied diagonal KLA and angle-sign/gauge unit tests;
6. FSDP construction, buffer dtype, state-dict, save/resume, and rank-consistency tests;
7. multi-step health checks covering bootstrap-zero angle, subsequently live clock gradients, token variation, and no nonfinite/endpoints;
8. long-sequence and long-run soak tests beyond the step-520 design screen;
9. peak-memory and end-to-end throughput measurements against the same recipe, including the cost of the materialized FP32 clock tensor; and
10. the full downstream evaluation suite after a naturally completed training run.

A Reproducibility audit

RDiag implementation baseline. The stable implementation statements in this note were audited against Git commit `2ff0a4987a1b40f627fbd3efd1fb16ff0490e307` (tree `2fd5560d8337523383c20dda503eb0098f996dbc`). Relevant immutable commit/path regions are:

- `lit_gpt/rdiag_kla.py`, construction and buffers around lines 137–193, phase construction around 222–293, and the fused layer route around 409–516;

- `lit_gpt/kla_ops/rdiag_clock.py`, anchor and eager oracle around 72–119 and 190–270, fused clock around 283–388, and custom backward around 621–722;
- `lit_gpt/kla_ops/rdiag_direct.py`, precise clock softplus around 43–47, direct radial transform around 300–377, and isolated radial VJP beginning around 702;
- `lit_gpt/kla_ops/rdiag_phase.py`, negative scan contract and reference around 1–19 and 294–328;
- `lit_gpt/kla_ops/rdiag_bounded_phase.py`, bounded fail-visible phase contract around 1–11 and 41–172;
- `lit_gpt/kla_ops/rdiag_norm.py`, public fused normalization/rotation contract around 10–90;
- `lit_gpt/kla_ops/rdiag_norm_reference.py`, normalization, BF16 boundary, and adjacent-pair rotation around 24–96;
- `lit_gpt/kla_ops/diag_naive.py`, KLA equations around 14–32, 144–199, and 451–491;
- `lit_gpt/diag_kla.py`, process-noise and isotropic-output initialization around 280–300;
- `lit_gpt/config.py`, bounded RDiag recipe fields around 349–369 and 742–763;
- `lit_gpt/rdiag_training_health.py`, health thresholds and observed counters around 24–37 and 121–255, and fail-closed decision logic around 579–612;
- `sanboxes/kernels/rotational_diag_kla_v1/DESIGN.md`, moving-frame semantics and the pair-tied literal model; and
- `sanboxes/kernels/rotational_diag_kla_v1/RESULTS.md`, the correctness and v1 performance evidence around lines 128–166.

At that commit the source SHA-256 values, in the same order as the list above, are:

```
8f09aa5881681ad9fa958f109f0ebf07240bad7b05877376f7c1f76633b9879
b15293c665f941002525ce6ccbf346265c13a83bb4ca1e2f4a0a276289c8bd0c
95b4309cd07d3e55507b4be623769383451cbf0e72f1624a038c985dba3b1ac4
647dda58e7855f3139c460afab7db97242cc18dfd5c5dc29e44b4497bda68f30
8918f21b3d326e3387001595b709a54b30b0ad944a8760b4d83d4cbd07e3b49a
261489c0076bc4752fa909319917fd14037471dea5416300d34cb1319d35c1b8
0f9ad9598188979f3eab6e3b3872de109409a358751198f664d4ad5717b6393e
431be42bf1d8903a675d8ca4a6ff06afdda86516f77d2258f0aa556747ccee86
fbacfa199a38ac4fe58c18612e3a0971bb0acdcd6cb7dd126b4bc85caabfb8b3
84634c19916fa88bf0e4b7849d4e71e8984fe3e7f385bfc0ada726a5b7eaf02c
e352c78a009b0b24f0290429c16c5213e004b61aeec4fc0a3eb2a7636918a910
c8004393867cdc7c638117a444db74030e7abfabe55c79f795468d27e8561d4c
7f9a75e0237cb1c225da7734a2cbc903257687e368e4b675f8826d5889c072dc
```

The exact-mean follow-up is commit `fe1b1258f601092ab29ae3431ec5a703d7087b25` (tree `bade1969461e68b8a987343ead8cc0f8634ac9e4`). In that commit, `lit_gpt/kla_ops/rdiag_direct.py` has SHA-256

```
7252664763c8d3b3b7af424b194f210424896f4d43f92ba39ef52a38a11f5628
```

The precise softplus is around lines 43–47, FP32 clock materialization around 299–375, and the framework mean around 1155–1197. This is the frozen source for the `materialize-plus-torch.mean` contract in Eq. (36). H200 job 1777082 subsequently completed with 17 passed and 3 failed tests. The exact radial-clock bitwise tests passed for $K = 64$ and $K = 128$, including threshold/extreme inputs; fused operator parity, backward-association checks, finite full-layer checks, and every legacy

performance gate also passed. Each of the three sealed-reference full-layer comparisons stopped at its first mismatch, `layer.rdiag_angle_proj.weight`, for the $(H, D, K) = (16, 1024, 64)$, $(16, 2048, 128)$, and $(18, 2304, 128)$ geometries; later VJPs were not reached. Thus exact-mean operator qualification is positive, but full-layer sealed-reference parity—and therefore production qualification—remains open.

Installed Mamba implementation. The source audit used `mamba-ssm 2.3.2.post1` installed under the active `hdn` environment. Paths and SHA-256 identities are:

- `mamba_ssm/modules/mamba2.py`,
605e4439ff0baec8d8acaf4a191d9f0570eea9900065a065909124c472b08707; inspect roughly lines 95–136, 210–260, and 309–318;
- `mamba_ssm/modules/mamba3.py`,
de1b104865c941eb6eeac63c3a75d5913fc2e7a119d52991590ef19c381b05ac; inspect roughly lines 76–87, 150–178, and 220–237;
- `mamba_ssm/ops/triton/mamba3/angle_dt.py`,
7d99f8cff37371cfbfe9d4dd36f17a16135fb07f2c6e14add3ad4dd5c0034e2e; inspect roughly lines 83–118 and 298–337;
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_combined.py`,
410ad57555b5841fba7bd76fdafdc3ddf82741176489c3e3493f5c6f3c0c6012; inspect roughly lines 90–105 and 247–259; and
- `mamba_ssm/ops/triton/mamba3/mamba3_siso_fwd.py`,
68d8d0dba0e3913f2e996862603289fd0d7157e7dc57c872e04234be8bb82695; inspect roughly lines 256–332, 348–352, and 386–425.

The hashes, package version, and paths are the durable identity; line numbers are navigation hints.

Qualification evidence and present status. The lower-tail rejection is recorded in `/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/probe_v1_1774423/probe_trace.json`. The redesigned E2 decision is recorded in `/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/qualification_bounded_de_20260828_1145/D_E_DECISION.json`. The precision experiment is recorded in `/lustre-storage/checkpoints/ngocbh/hdn/diagnostics/rdiag_bounded_clock/radial_clock_precision_1776573/result.json`; focused VJP evidence is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_vjp_diag_k64_1776509.log`, and the five-case composed check is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_angle_order_1776788.log`. These artifacts support the specific observations in Section 8. The completed exact-mean H200 result is in `/checkpoints/ngocbh/hdn/ckpts/logs/rdiag_clock_exact_1777082.log`. These results do not close the production checklist above.